Practice questions 28 October 2026

Use $\pi = \dfrac{22}{7}$. For every question without a picture, first draw a rough diagram and mark all the given measurements on it. Show all your steps.

Purpose. This set revises the whole chapter. The questions follow the style of the NCERT textbook *Ganita Manjari*, Grade 9, Chapter 6. They go from easier to harder.

Competency focus. Students work backwards from a result, find and explain a mistake, test a claim, compare two methods, and apply results to real situations. Several questions ask students to draw a diagram.

Concepts assessed. Circumference of a circle (Q1, 12); the number π (Q2); arc length (Q3, 13); the stagger on a running track (Q4); area of a parallelogram (Q5, 7); area of a triangle and triangles between the same parallels (Q6); the median of a triangle (Q7); Heron's formula (Q8, 9); Brahmagupta's formula (Q10); Baudhāyana's method of squaring a rectangle (Q11); area of a circle (Q12, 15); area of a sector and a ring sector (Q13); area of a segment (Q14); composite figures (Q15).

Answers. Answers appear in grey after each question, with a rough answer diagram.

---

1. A circular lotus pond in a temple garden is fenced once with 132 m of wire. Draw a rough diagram.

(a) Find the radius of the pond.

(b) A second fence is put up 7 m outside the first fence, all the way round. Find how much more wire the second fence needs than the first.

(c) Another pond has a radius of 35 m. It also gets a second fence 7 m outside the first one. Will the extra wire needed be more than, less than, or the same as your answer to (b)? Explain.

Answer.

(a) The wire is the circumference of the pond, so $2 \times \dfrac{22}{7} \times r = 132$.

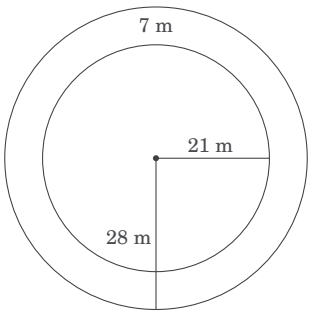
Therefore $r = 132 \times \dfrac{7}{2 \times 22} = 21$. The radius of the pond is 21 m.

(b) The second fence has radius $21 + 7 = 28$ m, so its circumference is $2 \times \dfrac{22}{7} \times 28 = 176$ m.

The extra wire is $176 - 132 = 44$ m.

(c) The extra wire is the same, 44 m. For any radius R the extra is $2 \times \dfrac{22}{7} \times (R+7) - 2 \times \dfrac{22}{7} \times R = 2 \times \dfrac{22}{7} \times 7 = 44$ m.

Check with $R = 35$: $2 \times \dfrac{22}{7} \times 42 - 2 \times \dfrac{22}{7} \times 35 = 264 - 220 = 44$ m.



2. Say whether each statement is true or false. Give a reason for each answer.

(a) The value of π is exactly $\frac{22}{7}$.

(b) For every circle, the circumference divided by the diameter gives the same number.

(c) Arjun measures a bangle. Its circumference is 22 cm and its diameter is 7 cm. Arjun says, "This proves that π is exactly $\frac{22}{7}$."

Answer.

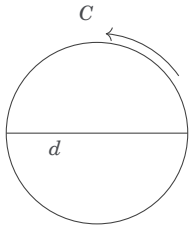
(a) False. π is irrational, so no fraction of two whole numbers equals it.

$\frac{22}{7} = 3.142\ldots$ and $\pi = 3.14159\ldots$, so $\frac{22}{7}$ is only a close approximation.

(b) True. This constant ratio is the same for every circle, and it is the number π.

(c) False. Every measurement is approximate, so a measurement cannot prove an exact value.

Also π is irrational, so π cannot equal $\frac{22}{7}$.



$\frac{C}{d} = \pi$ for every circle

3. A railway line goes round a curve. The curve is an arc of a circle of radius 210 m.

(a) A train goes round the curve and turns through an angle of 60°. Draw a rough diagram. Find the distance the train travels on the curve.

(b) On another curve of the same radius, a train travels 330 m. Find the angle through which the train turns.

Answer.

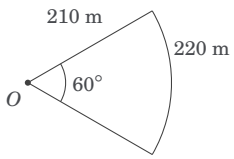
(a) The diagram below shows the arc, its radius and the angle at the centre.

The arc is the fraction $\frac{60}{360} = \frac{1}{6}$ of the whole circle.

Arc $= \frac{60}{360} \times 2 \times \frac{22}{7} \times 210 = \frac{1}{6} \times 1320 = 220$ m. The train travels 220 m.

(b) The whole circumference is $2 \times \frac{22}{7} \times 210 = 1320$ m.

The arc is $\frac{330}{1320} = \frac{1}{4}$ of the circle, so the angle is $\frac{1}{4} \times 360 = 90°$.



4. In a 200 m race on a school track, each runner runs one bend and one straight. The bend in every lane is a semicircle. Each lane is 1.4 m wide. All runners finish at the same line. Assume that each runner runs along the inner edge of the lane. Draw a rough diagram of lanes 1 and 2.

(a) Explain why the runner in lane 2 must start ahead of the runner in lane 1.

(b) Find how far ahead of the lane 1 runner the lane 2 runner must start.

(c) Find how far ahead of the lane 1 runner the lane 4 runner must start.

(d) Ravi says, "If the track had a bigger bend, the answer to (b) would change." Is Ravi right? Explain.

Answer.

(a) The two straights are equal, but the bend in lane 2 has the larger radius, so it is longer.

Therefore the lane 2 runner starts ahead, so that every runner covers 200 m.

(b) The radius of the lane 2 bend is $r + 1.4$, so the extra length is $\frac{22}{7} \times (r + 1.4) - \frac{22}{7} \times r = \frac{22}{7} \times 1.4 = 4.4$ m.

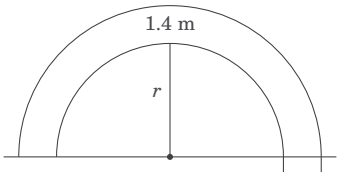
The lane 2 runner starts 4.4 m ahead of the lane 1 runner.

(c) The lane 4 bend has radius $r + 3 \times 1.4 = r + 4.2$, so the extra length is $\frac{22}{7} \times 4.2 = 13.2$ m, which is 3×4.4 m.

The lane 4 runner starts 13.2 m ahead of the lane 1 runner.

(d) No. The difference is $\frac{22}{7} \times 1.4 = 4.4$ m, and the radius r cancels from the subtraction.

So the stagger does not depend on the size of the bend, and Ravi is wrong.



5. A garden bed ABCD is shaped like a parallelogram. AB = 18 m and AD = 12 m. The perpendicular distance between the sides AB and DC is 10 m.

(a) Draw a rough diagram. Mark all the lengths and show the 10 m distance with a dotted line.

(b) Raju finds the area like this: Area $= 18 \times 12 = 216$ m^2. Explain Raju's mistake. Find the correct area of the bed.

(c) Find the perpendicular distance between the sides AD and BC.

Answer.

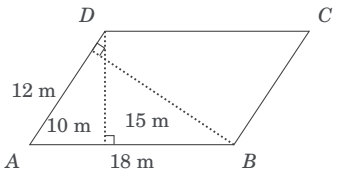
(a) See the diagram below. The base $AB = 18$ m and the slanting side $AD = 12$ m are marked, and the height of 10 m on AB is drawn dotted.

(b) Raju multiplied the two sides. The area of a parallelogram is base × perpendicular height on that base, and 12 m is a slanting side, not the height on AB.

Area $= 18 \times 10 = 180$ m^2.

(c) The area also equals $AD \times$ the perpendicular height on AD, so $12 \times h = 180$.

Therefore $h = 15$ m. The distance between AD and BC is 15 m.



6. A strip of cloth has two straight parallel edges that are 7 cm apart. Kavi draws two triangles, PQR and SQR, on the strip. Both triangles have the side $QR = 12$ cm along the lower edge. The points P and S are on the upper edge. In triangle PQR, the angles at Q and R are acute. In triangle SQR, the angle at Q is obtuse.

(a) Draw both triangles in one rough diagram. Show the height of each triangle with a dotted line.

(b) Find the area of each triangle.

(c) Priya says, "Triangle SQR has the larger area, because its sides are longer." Is Priya right? Explain.

Answer.

(a) See the diagram.

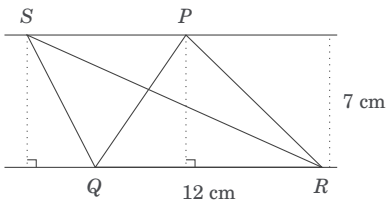
(b) Each triangle has base $QR = 12$ cm and height 7 cm, the distance between the parallel edges.

Area $= \dfrac{1}{2} \times 12 \times 7 = 42$ cm^2 each.

For SQR the height falls outside the triangle, on RQ extended, but it is still 7 cm.

(c) No. The area depends on the base and the perpendicular height, not on the slanting sides.

Both triangles have base 12 cm and height 7 cm, so both areas are 42 cm^2.



7. A glass panel $ABCD$ in a lamp is shaped like a parallelogram, and its area is 96 cm^2. Two thin metal strips run along the diagonals AC and BD. They meet at O and divide the glass into four triangular pieces. Draw a rough diagram.

(a) Explain why triangles AOB and BOC have equal areas. (Hint: what is BO in triangle ABC?)

(b) Explain why all four pieces have equal areas. Find the area of each piece.

(c) Neha says, "The four pieces are congruent." Is Neha right? Explain.

Answer.

(a) The diagonals of a parallelogram bisect each other, so O is the midpoint of AC.

So BO is a median of triangle ABC. A median divides a triangle into two triangles of equal area, so the area of AOB equals the area of BOC.

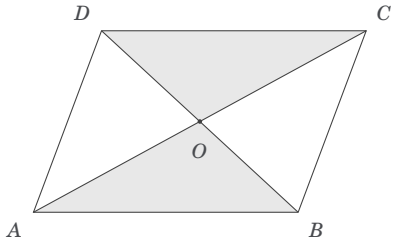
(b) In the same way, CO is a median of triangle BCD (O is the midpoint of BD), so the area of BOC equals the area of COD; and DO is a median of triangle CDA, so the area of COD equals the area of DOA.

So all four pieces are equal. Each $= 96 \div 4 = 24$ cm^2.

(c) Not in general. AOB is congruent to COD, and BOC is congruent to DOA (SAS: $OA = OC$, $OB = OD$, vertically opposite angles).

But AOB and BOC have different sides when $AB \neq BC$, so they are not congruent.

They are all congruent only when $ABCD$ is a rhombus. Equal area does not mean congruent.



8. A triangular plot ABC has sides $BC = 90$ m, $CA = 100$ m and $AB = 170$ m.

(a) Find the area of the plot using Heron's formula.

(b) Use your answer to (a) to find the shortest distance from the corner A to the line BC.

(c) Draw a rough diagram showing this shortest distance AD. Does D lie on the side BC, or on BC extended beyond C? Find CD and check your answer to (b) using AB.

Answer.

(a) $s = \dfrac{90 + 100 + 170}{2} = 180$. $s - a = 90$, $s - b = 80$, $s - c = 10$.

Area $= \sqrt{180 \times 90 \times 80 \times 10} = \sqrt{12960000} = 3600$ m^2.

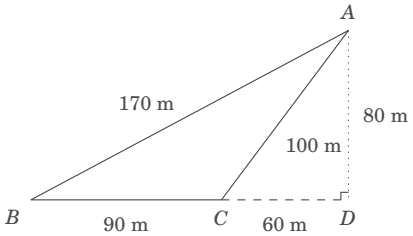
(b) $\dfrac{1}{2} \times 90 \times AD = 3600$, so $AD = 80$ m.

(c) Angle C is obtuse, so D lies on BC extended beyond C.

In right triangle ADC: $CD = \sqrt{100^2 - 80^2} = \sqrt{3600} = 60$ m. $BD = 90 + 60 = 150$ m.

In right triangle ADB: $AB = \sqrt{150^2 + 80^2} = \sqrt{28900} = 170$ m.

This matches, so $AD = 80$ m is correct.



9. A garden bed is shaped like an isosceles triangle. Its perimeter is 64 m and its unequal side (the base) is 24 m. Draw a rough diagram.

(a) Find the length of each equal side.

(b) Find the area of the bed using Heron's formula.

(c) Find the area again in a different way, using the height on the base. Check that both answers agree.

Answer.

(a) $(64 - 24) \div 2 = 20$ m.

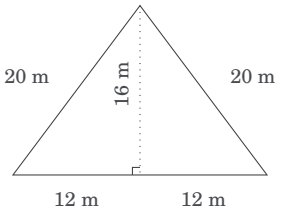
(b) $s = 32$; $s - a = 12$, $s - b = 12$, $s - c = 8$.

Area $= \sqrt{32 \times 12 \times 12 \times 8} = \sqrt{36864} = 192$ m^2.

(c) The height from the top corner meets the base at its midpoint, 12 m from each end.

Height $= \sqrt{20^2 - 12^2} = \sqrt{256} = 16$ m. Area $= \dfrac{1}{2} \times 24 \times 16 = 192$ m^2.

Both answers agree.

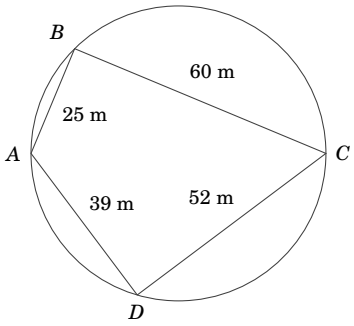


10. A plot of land $ABCD$ has all four corners on a circle, as shown. $AB = 25$ m, $BC = 60$ m, $CD = 52$ m and $DA = 39$ m.

(a) Find the area of the plot using Brahmagupta's formula.

(b) The diagonal AC is a diameter of the circle, and $AC = 65$ m. So the angles at B and D are right angles. Find the area of the plot again by splitting it along AC into two triangles. Check that it agrees with (a).

(c) A rectangle 12 m by 5 m also has all four corners on a circle. Use Brahmagupta's formula to find its area. Does the formula give the answer you expect? Explain.



Answer.

(a) $s = \dfrac{25+60+52+39}{2} = 88$. $s-a = 63$, $s-b = 28$, $s-c = 36$, $s-d = 49$.

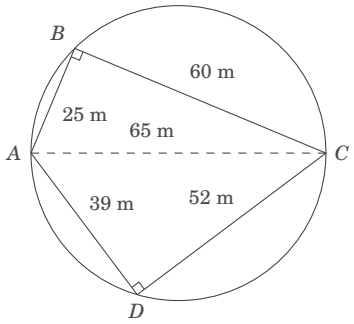
Area $= \sqrt{63 \times 28 \times 36 \times 49} = \sqrt{3111696} = 1764$ m^2.

(b) Area of $ABC = \dfrac{1}{2} \times 25 \times 60 = 750$ m^2; area of $ACD = \dfrac{1}{2} \times 39 \times 52 = 1014$ m^2; total $= 750 + 1014 = 1764$ m^2.

This agrees with (a).

(c) $s = \dfrac{12+5+12+5}{2} = 17$; $(17-12)(17-5)(17-12)(17-5) = 5 \times 12 \times 5 \times 12 = 3600$; area $= \sqrt{3600} = 60$ m$^2 = 12 \times 5$.

Yes: for a rectangle the formula gives $\sqrt{b \times l \times b \times l} = l \times b$, the correct area.



11. Long ago, Baudhāyana showed how to change a rectangle into a square of the same area. Follow his steps for a rectangular field 18 m long and 8 m wide. Draw a diagram for each step.

(a) Cut off a square of side 8 m from one end of the rectangle. What are the length and breadth of the strip that is left?

(b) Cut the strip into two equal strips, each 8 m long. Place one strip along the right side of the square and the other along the top of the square. This makes an L-shape. Show that the L-shape fits inside a square of side 13 m, with a small square missing at one corner. Find the side of the missing square.

(c) Find the area of the L-shape in two ways: from the rectangle, and as (big square) − (small square).

(d) Baudhāyana now draws a right triangle with hypotenuse 13 m and one side 5 m. Find the third side. Explain why a square on this third side has the same area as the field.

Answer.

(a) The square cut off has side 8 m, so the strip left is $18 - 8 = 10$ m long and 8 m broad.

Its area is $10 \times 8 = 80$ m^2.

(b) Each new strip is 5 m by 8 m, because the 10 m side of the strip is cut in half.

The square then measures $8 + 5 = 13$ m along each side, except at the corner where a 5 m by 5 m square is empty.

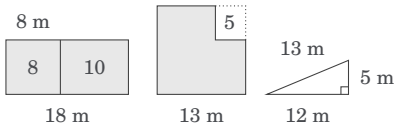
So the side of the missing square is 5 m.

(c) From the rectangle: $18 \times 8 = 144$ m^2.

From the squares: $13^2 - 5^2 = 169 - 25 = 144$ m^2. Both methods give 144 m^2.

(d) The third side is $\sqrt{13^2 - 5^2} = \sqrt{144} = 12$ m.

By Pythagoras, the square on this side has area $12^2 = 144$ m^2, and $13^2 - 5^2$ is also the area of the L-shape, which equals the area of the field.



12. The top of a round table has a circumference of 220 cm.

(a) Find the radius of the table top.

(b) Find the area of the table top.

(c) A bigger round table has a radius twice as long. Ravi says, "Its top has twice the area." Is Ravi right? Find the area of the bigger table top and explain.

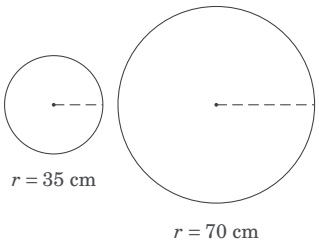
Answer.

(a) $2 \times \frac{22}{7} \times r = 220$, so $r = 220 \times \frac{7}{22} \times \frac{1}{2} = 35$ cm.

(b) Area $= \frac{22}{7} \times 35 \times 35 = 3850$ cm^2.

(c) No. The new radius is 70 cm, so its area is $\frac{22}{7} \times 70 \times 70 = 15400$ cm^2, which is 4×3850 cm^2.

When the radius doubles, $r \times r$ becomes four times as large, so the area becomes four times as large, not twice.



13. A car's wiper arm is 21 cm long and turns about one end, O. The rubber blade is fixed to the outer part of the arm, from 7 cm to 21 cm from O. The arm turns through an angle of 90°.

(a) Draw a rough diagram. Shade the part of the glass that the blade wipes.

(b) Find the area of the glass that the blade wipes.

(c) Find the length of the path of the tip of the arm.

Answer.

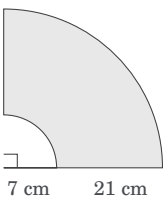
(a) See the diagram.

(b) The wiped part is the difference of two sectors of angle 90°, with radii 21 cm and 7 cm.

Area $= \frac{90}{360} \times \frac{22}{7} \times (21 \times 21 - 7 \times 7) = \frac{1}{4} \times \frac{22}{7} \times 392 = 308$ cm^2.

(c) The tip moves along an arc of radius 21 cm.

Path $= \frac{90}{360} \times 2 \times \frac{22}{7} \times 21 = 33$ cm.

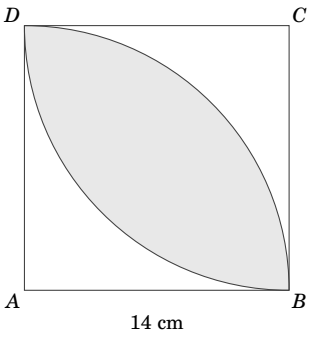


14. A floor tile $ABCD$ is a square of side 14 cm. Two quarter circles are drawn inside it: one with centre A and radius AB, and one with centre C and radius CB. Both arcs go from B to D. The leaf-shaped part between the two arcs is shaded.

(a) Find the area of the shaded leaf.

(b) Find the perimeter of the shaded leaf.

(c) Meena finds the area of the leaf like this: (area of one quarter circle) + (area of the other quarter circle) − (area of the square). Explain why her method works, and check that it gives the same answer.



Answer.

(a) The diagonal BD cuts the leaf into two equal segments.

One segment $= \frac{1}{4} \times \frac{22}{7} \times 14 \times 14 - \frac{1}{2} \times 14 \times 14 = 154 - 98 = 56$ cm^2.

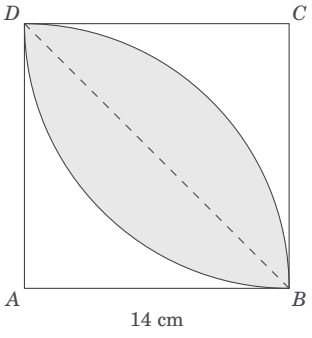
Leaf $= 2 \times 56 = 112$ cm^2.

(b) Each arc is a quarter of a circle of radius 14 cm.

Arc $= \frac{1}{4} \times 2 \times \frac{22}{7} \times 14 = 22$ cm, so the perimeter $= 2 \times 22 = 44$ cm.

(c) The two quarter circles together cover the whole square, and they overlap exactly in the leaf.

So $154 + 154 = 196 +$ leaf, and the leaf $= 308 - 196 = 112$ cm^2, which is the same answer.



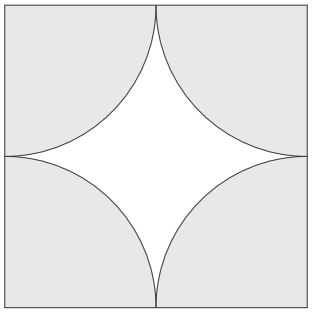
15. A square garden has side 28 m. At each corner there is a flower bed shaped like a quarter circle of radius 14 m, with its centre at the corner of the garden. The rest of the garden is lawn.

(a) Find the area of the lawn.

(b) Find the perimeter of the lawn.

(c) The gardener makes the flower beds smaller, with radius 7 m. Find the new area of the lawn and the new perimeter of the lawn.

(d) Did the perimeter of the lawn increase or decrease in (c)? Explain why.



Answer.

(a) The four quarter circles make one full circle of radius 14 m.

Beds $= \frac{22}{7} \times 14 \times 14 = 616$ m^2, so the lawn $= 28 \times 28 - 616 = 784 - 616 = 168$ m^2.

(b) The boundary of the lawn is made of four quarter arcs, which make one full circumference.

Perimeter $= 2 \times \frac{22}{7} \times 14 = 88$ m.

(c) With radius 7 m the beds make one circle of area $\frac{22}{7} \times 7 \times 7 = 154$ m^2.

Lawn $= 784 - 154 = 630$ m^2. Perimeter $= 2 \times \frac{22}{7} \times 7 = 44$ m of arcs, plus four straight pieces of $28 - 7 - 7 = 14$ m each, so $44 + 4 \times 14 = 100$ m.

(d) The perimeter increased, because 100 m is more than 88 m.

With radius 14 m the beds touch at the midpoints of the sides, so the boundary of the lawn is only arcs. With radius 7 m the arcs are shorter, but four straight pieces appear along the sides of the square.

